

Forgotten Polygons: Multimodal Large Language Models are Shape-Blind

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Abstract

Despite strong performance on vision-language tasks, Multimodal Large Language Models (MLLMs) struggle with mathematical problem-solving, with both open-source and state-of-the-art models falling short of human performance on visual-math benchmarks. To systematically examine visual-mathematical reasoning in MLLMs, we (1) evaluate their understanding of geometric primitives, (2) test multi-step reasoning, and (3) explore a potential solution to improve visual reasoning capabilities. Our findings reveal fundamental shortcomings in shape recognition, with top models achieving under 50% accuracy in identifying regular polygons. We analyze these failures through the lens of dual-process theory and show that MLLMs rely on System 1 (intuitive, memorized associations) rather than System 2 (deliberate reasoning). Consequently, MLLMs fail to count the sides of both familiar and novel shapes, suggesting they have neither learned the concept of “sides” nor effectively process visual inputs. Finally, we propose Visually Cued Chain-of-Thought (VC-CoT) prompting, which enhances multi-step mathematical reasoning by explicitly referencing visual annotations in diagrams, boosting GPT-4o’s accuracy on an irregular polygon side-counting task from 7% to 93%. Our findings suggest that System 2 reasoning in MLLMs remains an open problem, and visually-guided prompting is essential for successfully engaging visual reasoning. Code available at: <https://github.com/rsinghlab/Shape-Blind> *.

1 Introduction

As Multimodal Large Language Models (MLLMs) demonstrate success in vision-language tasks (Deitke et al., 2024; Chen et al., 2025), researchers seek to analyze their underlying mechanisms (Golovanevsky et al., 2025; Jiang et al., 2024; Luo et al.,

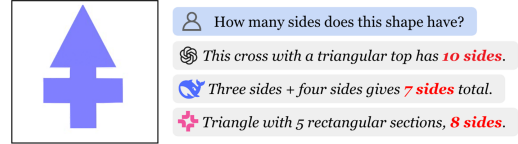


Figure 1: Multimodal Large Language Models (MLLMs), such as GPT-4o, Janus-Pro, and Molmo fail at counting the number of sides of novel shapes.

2024) and evaluate their ability to perform generalized problem-solving compared to human reasoning (Cherian et al., 2024). A key aspect of human reasoning is the interplay between intuitive, reflexive thinking (known as System 1) and deliberate, logical reasoning (known as System 2) (Evans, 2008; Evans and Stanovich, 2013; Kahneman, 2011). While humans engage in both types of reasoning, a key question arises: *Can MLLMs move beyond memorized responses of System 1 to engage in the analytical reasoning of System 2?*

Mathematics provides a powerful framework for this investigation, as fundamental reasoning skills are crucial for complex problem-solving. The combination of logic, abstract concepts, and specialized symbolic language makes mathematical reasoning problems a strong test of the extent to which models engage in System 2 reasoning rather than relying solely on memorized concepts (Satpute et al., 2024). Studying System 2 reasoning abilities is especially relevant given recent progress in natural language processing, where Large Language Models (LLMs) like GPT-4 (Achiam et al., 2023), Qwen (Bai et al., 2023), and LLaMA (Touvron et al., 2023) have demonstrated notable success on mathematical benchmarks like GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021). Despite these advancements, MLLMs remain limited in visual-mathematical reasoning. While closed-source models like GPT-4o demonstrate potential, their performance on benchmarks such as

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MathVerse (Zhang et al., 2024a) remains far below human levels, with open-source models typically performing even worse (Zhang et al., 2024a). Even MLLMs specially trained for visual, mathematical reasoning (G-LLaVA (Gao et al., 2023), Math-PUMA (Zhuang et al., 2024), Math-LLaVa (Shi et al., 2024) cannot close the gap between open-source and closed-source models. Many benchmarks feature complex geometric diagrams that require multi-step reasoning, with human performance reaching no more than 70% accuracy (Wang et al., 2024a; Zhang et al., 2024a). Cherian et al. (2024) show that MLLMs tend to perform better on more complex mathematical tasks but struggle with simple tasks designed for young children. This counterintuitive result raises concerns about whether MLLM performance on visual-mathematical benchmarks reflects genuine mathematical reasoning (System 2) or merely the retrieval of familiar concepts from training data (System 1).

We design a two-part task to evaluate MLLMs’ reasoning abilities: (1) shape recognition, which relies on visual recall (System 1), and (2) side counting, which engages visual reasoning (System 2). Starting with common geometric shapes and properties allows us to pinpoint specific failures in how MLLMs process mathematical diagrams. Our results show that while the underlying LLMs demonstrate perfect accuracy on non-visual questions about polygon names and side counts, their multimodal counterparts fail when models have to rely on images to answer questions correctly. Unlike humans, who can count a shape’s sides and infer its identity (e.g., “1, 2, . . . , 7” $\rightarrow$ “heptagon”), our results suggest that MLLMs do not attempt such reasoning. Instead, they rely on System 1 heuristics like shape memorization and spurious correlations between visual features and labels, ignoring vital information in the image.

To understand why MLLMs struggle with polygon identification from visual inputs, we analyze vision encoder embeddings. Our results show that vision encoders are “shape-blind”: common shapes form distinct clusters, while less frequent ones like pentagons, heptagons, and octagons overlap (Section 3.1.3). Even models specifically trained for geometric understanding, such as G-LLaVA, Math-LLaVa, and Math-PUMA, exhibit shape-blindness where distinct polygons are embedded in the same region in vector space. We further demonstrate vision encoders’ shape-blindness with a two-shape multi-step task: identifying shapes, counting the

sides, and computing the total sum of sides, adding complexity by incorporating arithmetic into the reasoning process. On average, models correctly identify both shapes in an image only 27.58% of the time, as shape recognition depends entirely on visual input. However, MLLMs compute the sum of the identified shapes’ sides correctly in 70.75% of cases, indicating that despite errors in recognition, the summation operation remains highly accurate. While open-source models struggle with less common polygons, GPT-4o identifies shapes with at least 49% accuracy. To test whether models use System 2 reasoning, we introduce a dataset of abstract shapes and irregular polygons unlikely to have appeared in training (see Figure 1). No model accurately counts the sides of these novel shapes, reinforcing their reliance on memorization rather than genuine geometric understanding.

As MLLMs continue to advance, our findings urge the community to re-examine the complexity of visual-mathematical benchmarks. If state-of-the-art open-source models fail to recognize simple shapes, should we be evaluating them on complex geometric tasks? Our results suggest that current models rely heavily on factual recall rather than true geometric reasoning, limiting their ability to generalize beyond familiar shapes. To explore ways of bridging this gap, we take a step forward with *Visually-Cued Chain-of-Thought* (VC-CoT) prompting, leveraging the fact that many geometric datasets already contain annotated shapes (e.g., labeled triangle vertices A, B, and C). By explicitly guiding the model to reference image annotations (letters or numbers) and reason about their relationships, VC-CoT significantly improves GPT-4o’s ability to count the sides of novel shapes, boosting accuracy from 7% to 93%. Across Molmo, Janus-Pro, GPT-4-Turbo, and GPT-4o, our VC-CoT prompts achieve an average accuracy improvement of approximately 5% compared to standard CoT prompts on MathVerse. These findings highlight the importance of structured prompting in strengthening the connection between visual perception and mathematical reasoning in MLLMs.

2 Related Works

Despite the success of MLLMs, emerging research highlights their struggles with reasoning. Alhamoud et al. (2025) find that vision-language models struggle to negate in retrieval and multiple-choice tasks. Beyond general reasoning, open-

source, closed-source, and math-specific MLLMs perform poorly on visual math benchmarks (Wang et al., 2024a; Zhang et al., 2024a).

A key limitation in MLLMs is the vision encoder. Tong et al. (2024) show CLIP-based vision encoders fail to capture fine-grained details — a detrimental trait for tasks requiring precise visual reasoning. Similarly, in the visual-mathematics domain, Gao et al. (2023); Shi et al. (2024); Zhang et al. (2024b); Zhuang et al. (2024) demonstrate that vision encoders produce inadequate representations of mathematical diagrams and vision-text misalignments hinder multimodal reasoning.

One way to address these challenges is fine-tuning. Recent works adopt a multi-step approach: (1) fine-tuning vision encoders to enhance visual representation, (2) training modality projectors for better alignment, and (3) instruction-tuning with CoT datasets (Gao et al., 2023; Shi et al., 2024; Zhang et al., 2024b; Zhuang et al., 2024). While this may increase performance on math benchmarks, studies show that fine-tuning foundation models often reduces generalization ability and does not address overarching reasoning capability (Yang et al., 2024; Chu et al., 2025).

In the language domain, Chain-of-Thought (CoT) prompting has proven highly effective in encouraging System 2 reasoning in LLMs (Wei et al., 2023). Xiang et al. (2025) further reinforces this, demonstrating that structuring reasoning through CoT can significantly improve logical inference and problem-solving. However, applying CoT to MLLMs has been far less successful. Zhang et al. (2024c) show that open-source MLLMs struggle with CoT, largely due to limitations in existing visual-instruction tuning datasets, which prioritize short, simplistic responses over structured reasoning. Thus, recent works explicitly fine-tune MLLMs for CoT reasoning, particularly in object counting and mathematical reasoning tasks (Zhang et al., 2024c,b; Zhuang et al., 2024; Deitke et al., 2024). However, even these fine-tuned models continue to struggle on complex visual-mathematical benchmarks such as MathVista (Lu et al., 2024), and MathVerse (Zhang et al., 2024a). These challenges highlight that despite recent efforts, System 2 reasoning in MLLMs remains an open problem, with current approaches failing to achieve generalizable reasoning abilities.

3 Experiments

Models Evaluated: In our experiments, we evaluate 13 diverse MLLMs. We consider (1) general open-source models, including LLaVA-1.5-7B (Liu et al., 2024), LLaVA-Next-7B (Li et al., 2024b), LLaVA-OneVision-7B (Li et al., 2024a), Qwen2-VL-7B (Wang et al., 2024b), InternVL-8B (Wang et al., 2024b), Molmo-7B (Deitke et al., 2024), and DeepSeek’s Janus Pro-7B (Chen et al., 2025); (2) math-specialized open-source models, including Math-LLaVA-13B (Shi et al., 2024), G-LLaVA-7B (Gao et al., 2023), and Math-PUMA-7B (Zhuang et al., 2024); and (3) closed-source models, including GPT-4-Turbo and GPT-4o. See Table 8 in Appendix A for full evaluation details.

3.1 Probing Geometric Knowledge

Geometric reasoning provides a natural testbed for distinguishing between System 1 heuristics, such as memorization of common shape patterns, and System 2 reasoning, which requires deliberate reasoning, such as counting sides of a shape. To examine this distinction, we create a dataset comprising six common regular polygons: triangle, square, pentagon, hexagon, heptagon, and octagon (see examples in Figure 6 in Appendix B.1). For each shape, we generate images with different colors, rotations, and sizes, creating a dataset of 2000 images. All images are 400×400 pixels, and we ensure that even the smallest shapes occupy at least 15% of the image (more details in Appendix Section B). Our analysis follows a three-stage approach to disentangle the contributions of vision, language, and their integration in MLLMs. First, we assess the MLLM as a whole, evaluating its ability to connect visual information to geometric properties (Section 3.1). Next, we prompt the underlying LLMs in a text-only setting to determine their knowledge independent of vision (Section 3.1.2). Finally, we analyze the vision encoder by examining the vision embeddings, shedding light on what the model “sees” without language guidance (Section 3.1.3).

For each image, we use two prompts to test for geometric knowledge: “What shape is in the image?” and “How many sides does the shape in the image have?”. Table 1 shows that MLLMs excel at identifying common shapes like triangles and squares, with many models achieving perfect accuracy. However, performance drops sharply for less familiar polygons such as pentagons, heptagons, and octagons. Most models fail on heptagons, with